Table of Common Distribution

Name	Notation	pdf/pmf	Range	Mean	Variance
Beta	$\text{Beta}(a,b)$	$f(x)=\frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)}x^{a-1}(1-x)^{b-1}$	$x\in(0,1)$	$\frac{a}{a+b}$	$\frac{ab}{(a+b)^2(a+b+1)}$
Bernoulli	$\text{Ber}(p)$	$f(x)=p^x(1-p)^{1-x}$	$x\in\{0,1\}$	p	$p(1-p)$
Binomial	$\text{Bin}(n,p)$	$f(x)=\binom{n}{x}p^x(1-p)^{n-x}$	$x\in\{0,1,..,n\}$	np	$np(1-p)$
Exponential	$\text{Exp}(\lambda)$	$f(x)=\lambda e^{-\lambda x}$	$x\in\mathbb{R}_+$	$1/\lambda$	$1/\lambda^2$
Gamma	$\text{Ga}\ (a,b)$	$f(x)=\frac{b^a}{\Gamma(a)}x^{a-1}e^{-bx}$	$x\in\mathbb{R}_+$	a/b	a/b^2
Normal	$N\left(\mu,\sigma^2\right)$	$f(x)=\frac{1}{\sqrt{2\pi\sigma^2}}e^{-\frac{(x-\mu)^2}{2\sigma^2}}$	$x\in\mathbb{R}$	μ	σ^2
Poisson	$\text{Pois}\ (\lambda)$	$f(x)=\frac{\lambda^x}{x!}e^{-\lambda}$	$x\in\{0,1,\ldots\}$	λ	λ
Uniform	$\text{Unif}\ (a,b)$	$f(x)=\frac{1}{b-a}$	$x\in(a,b)$	$(a+b)/2$	$(b-a)^2/12$

(The gamma function is defined as $\Gamma(x)=(x-1)!$ for integer x.)